

Soil Health & Crop Recommendation Report

Report Date: November 15, 2025 at 22:10:08 UTC

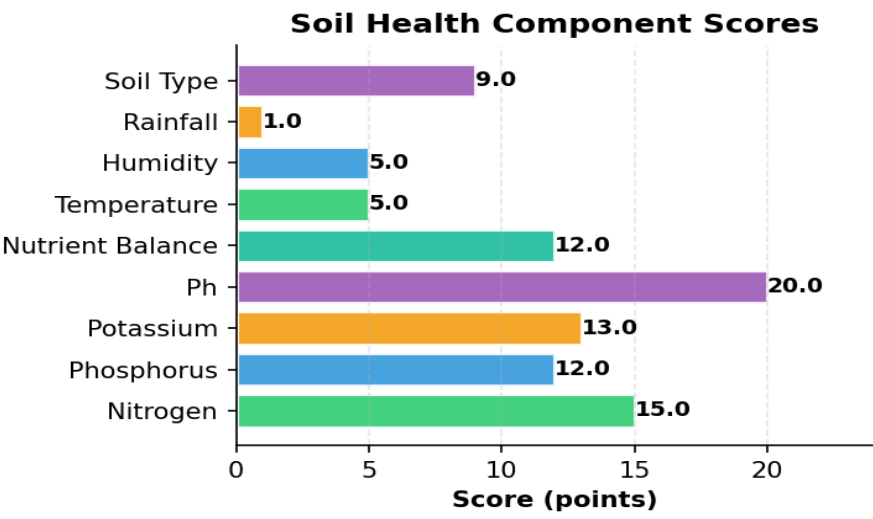
Farmer Information	
Name:	lasson
Phone:	8088117817
Location:	Vijayapura
Land Size:	1.00 acres

Parameter	Value	Unit
Nitrogen (N)	100.0	mg/kg
Phosphorus (P)	40.0	mg/kg
Potassium (K)	30.0	mg/kg
pH Level	6.50	
Temperature	26.5	°C
Humidity	60.0	%
Rainfall	900.0	mm
Soil Type	Black Soil	

Soil Health Assessment

Overall Soil Health Score	92.0/100	Excellent	
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Component	Score	Points
Nitrogen	15.0	points
Phosphorus	12.0	points
Potassium	13.0	points
Ph	20.0	points
Nutrient Balance	12.0	points
Temperature	5.0	points
Humidity	5.0	points
Rainfall	1.0	points
Soil Type	9.0	points



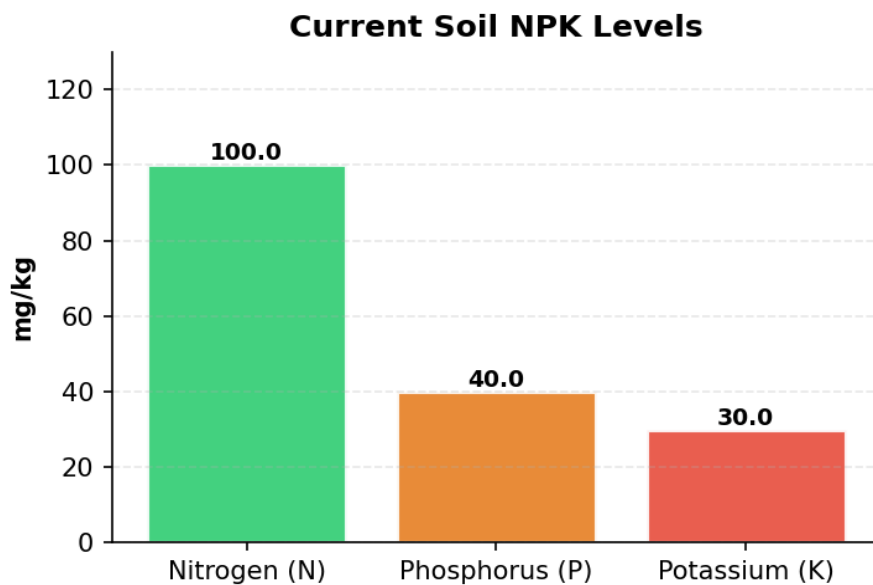
Recommended Crop

Predicted Crop	Cotton
Confidence Level	20.1%

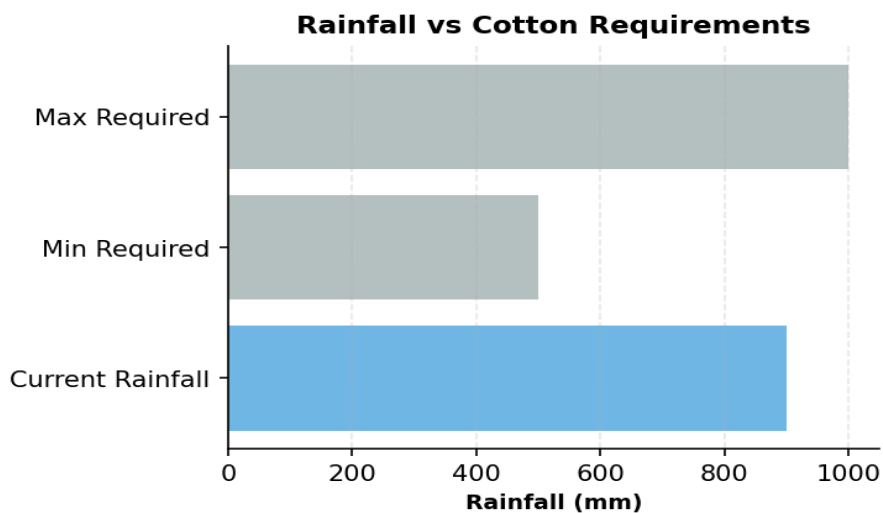
Alternative Crop Options

Rank	Crop	Confidence (%)
1	Pigeonpeas	13.9%
2	Jute	11.0%

Soil NPK Levels Visualization



Rainfall Analysis



Soil Treatment Recommendations

- Soil pH (6.5) is within optimal range (5.8-8.0) for cotton.
- Nitrogen (100.0 mg/kg) is within optimal range for cotton.
- Phosphorus (40.0 mg/kg) is within optimal range for cotton.
- Potassium (30.0 mg/kg) is low for cotton. Optimal range: 40-60. Add wood ash, banana peel powder, or potash (20 kg/acre needed).
- High clay content, retains moisture well. Ideal for cotton, wheat, and pulses.
- N:K ratio is high (>2:1). Consider balancing with potassium.

Fertilizer Recommendations

Fertilizer Requirements for cotton:

- Land Size: 1.0 acres
- Current NPK: N=100.0, P=40.0, K=30.0 mg/kg
- Ideal NPK: N=100, P=45, K=50 kg/acre

Required Application:

- Phosphorus (P): 5.0 kg total (5.0 kg/acre)
→ Apply as basal dose before sowing/transplanting
- Potassium (K): 20.0 kg total (20.0 kg/acre)
→ Apply in 2 splits: 50% basal, 50% at flowering

Organic Alternatives:

- Compost/FYM: 5-10 tons/acre (provides N, P, K)
- Green Manure: Leguminous crops (fixes N)
- Biofertilizers: Rhizobium (N), PSB (P), VAM (P)

Crop-Specific Tips for Cotton

Avoid waterlogging; monitor for pests like bollworms. Requires warm climate.

Additional Notes

- These recommendations are based on comprehensive soil analysis and advanced ML predictions.
- Actual results may vary based on local conditions, weather patterns, and farming practices.
- Consult with local agricultural experts before making major farming decisions.
- Regular soil testing (every 2-3 years) is recommended for optimal crop management.
- Consider crop rotation and organic farming practices for long-term soil health.
- Monitor weather forecasts and adjust irrigation schedules accordingly.

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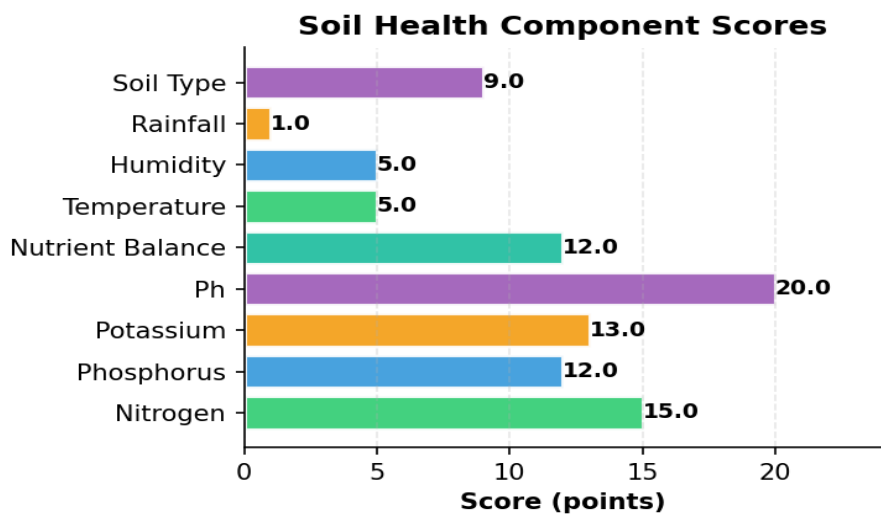
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Soil Health & Crop Recommendation System

Comprehensive Agricultural Analysis Report



Recommended Crop

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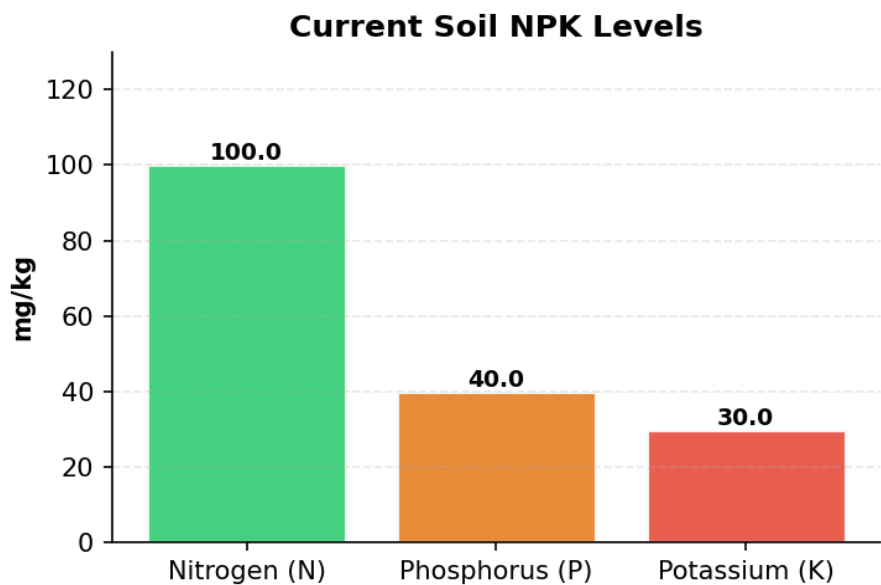
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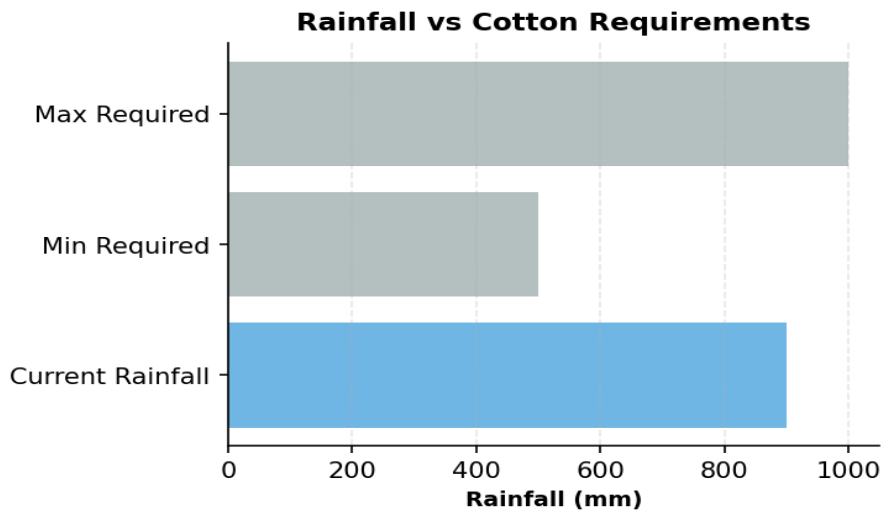
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